

Finding groups and patterns in large datasets

Clustering intuition and practical examples.

Suitable questions, preprocessing, cluster stability, and interpretation.



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Research
Question

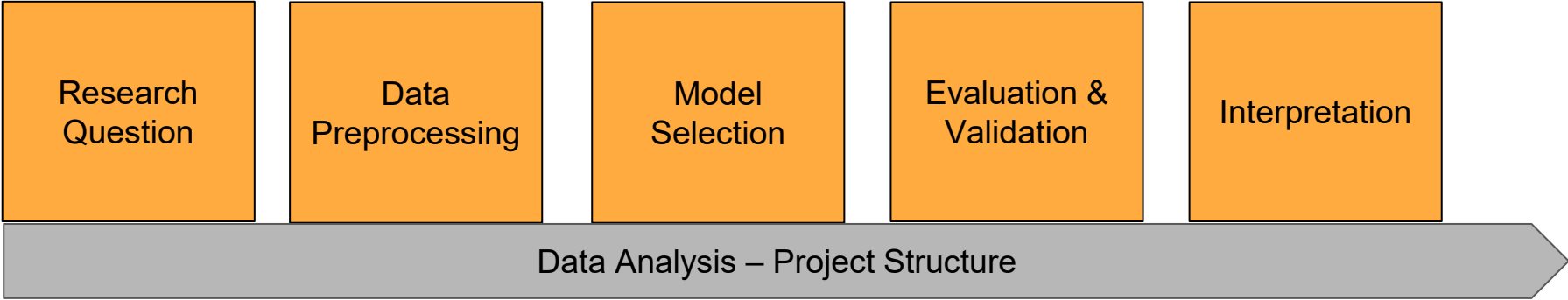
Data
Preprocessing

Model
Selection

Evaluation &
Validation

Interpretation

Data Analysis – Project Structure



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graph LR; A[Research Question] --> B[Data Preprocessing]; B --> C[Model Selection]; C --> D[Evaluation & Validation]; D --> E[Interpretation];
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Unsupervised Machine Learning

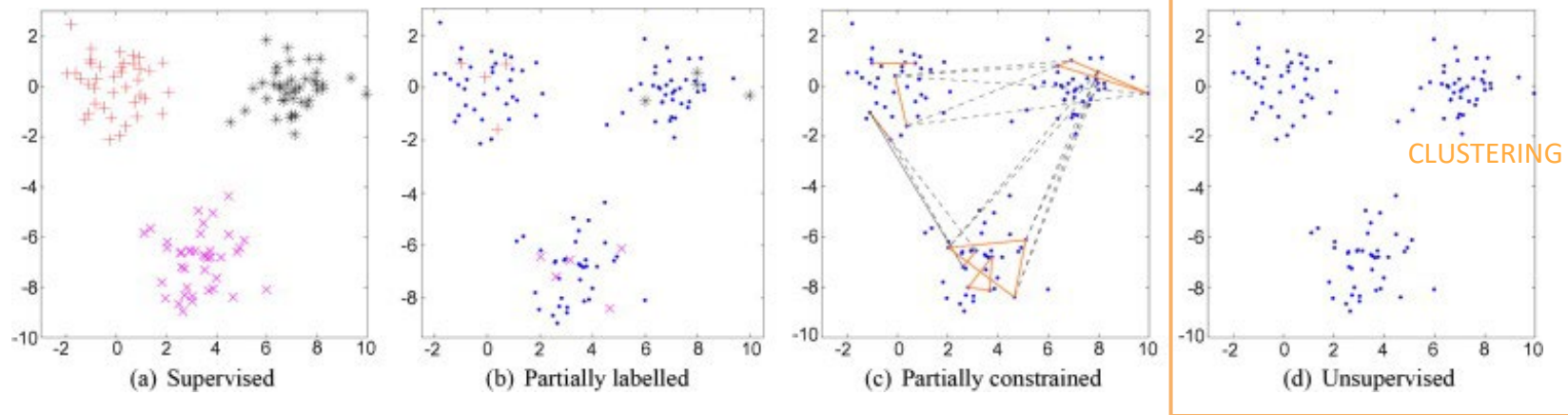
Population Segmentation

Multimorbidity-based Population Segmentation

Use Case: Clustering Multimorbidity on NFHS-5 India

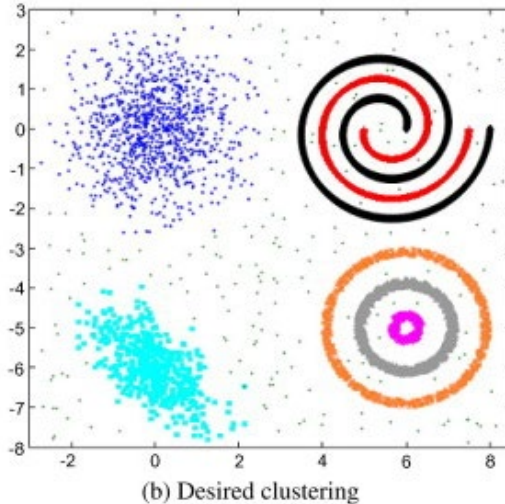
1. Research Question – find discretized embedding

Purpose of Unsupervised Machine Learning: Identification of underlying structures, natural classifications, or data compression = “Discretized embedding”



“Clustering is a learning task that aims to decompose a given set of observations into subgroups (clusters) based on data similarity, such that observations in the same cluster are more closely related to each other than observations in different clusters. It is an unsupervised learning task, since it identifies structures in unlabeled datasets, and a classification task, since it can give a label to observations according to the cluster they are assigned to.”

2. Data Preprocessing - Similarity & Dissimilarity Metrics



- seven clusters differ in terms of their shape, size, and density.
- presence of noise in the data makes the detection of the clusters even more difficult.

An ideal cluster :

„A cluster is a set of points that is compact and isolated.“

In reality:

„A cluster is a subjective entity that is in the eye of the beholder and whose significance and interpretation requires domain knowledge.“

But, while humans are excellent cluster seekers in two and possibly three dimensions, we need automatic algorithms for high-dimensional data.

Challenge one: Although these clusters are apparent to a data analyst, none of the available clustering algorithms can detect all these clusters. → importance of method selection

Challenge two: Number of clusters not known

2. Data Preprocessing - Similarity & Dissimilarity Metrics

- **similarity** is fundamental to the definition of a cluster and measures the similarity between two instances drawn from the same feature space
→ qualitative features
- **dissimilarity** (distance measure) = opposite measure to similarity, which is often calculated instead
→ quantitative features
- Metrics need to be chosen carefully, depending on:
 - Feature types and heterogeneity
 - Feature scales (normalization)
 - Feature correlations (Mahalanobis distance)
 - effect of surrounding or neighboring points
= **context**
 - additional domain information = **concept**

The most popular metric for continuous features is the **Euclidean distance**

$$d_2(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{k=1}^d (x_{i,k} - x_{j,k})^2 \right)^{1/2}$$

$$= \|\mathbf{x}_i - \mathbf{x}_j\|_2,$$

Might not always be the best choice (context dependent), e.g.:

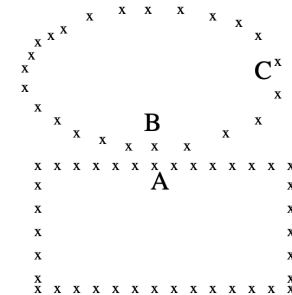
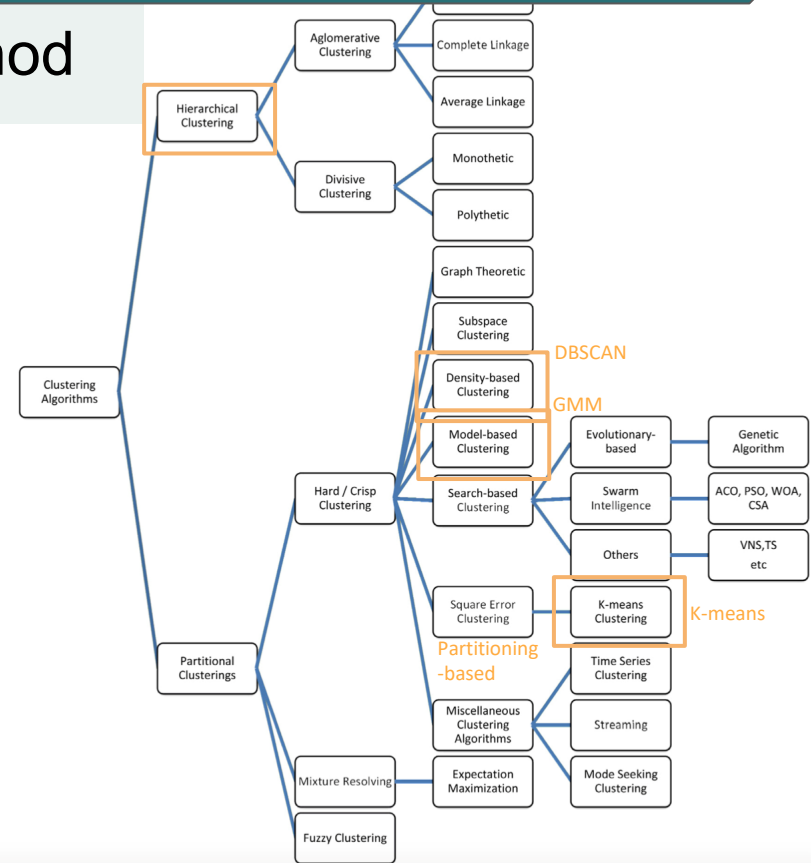


Figure 6. Conceptual similarity between points .

3. Model Selection - Clustering Method

Traditional algorithms

Category	Typical algorithm
Clustering algorithm based on partition	K-means, K-medoids, PAM, CLARA, CLARANS
Clustering algorithm based on hierarchy	BIRCH, CURE, ROCK, Chameleon
Clustering algorithm based on fuzzy theory	FCM, FCS, MM
Clustering algorithm based on distribution	DBCLASD, GMM
Clustering algorithm based on density	DBSCAN, OPTICS, Mean-shift
Clustering algorithm based on graph theory	CLICK, MST
Clustering algorithm based on grid	STING, CLIQUE
Clustering algorithm based on fractal theory	FC
Clustering algorithm based on model	COBWEB, GMM, SOM, ART



3. Clustering Method based on Hierarchy

- Goal = construct the hierarchical relationship among data in order to cluster
- Suppose that each data point stands for an individual cluster in the beginning, and then, the most neighboring two clusters are merged into a new cluster until there is only one cluster left. Or reverse...
- Advantages: suitable for arbitrary data shape and type, relatively high scalability
- Disadvantages: high in time complexity, the number of clusters needs to be preset

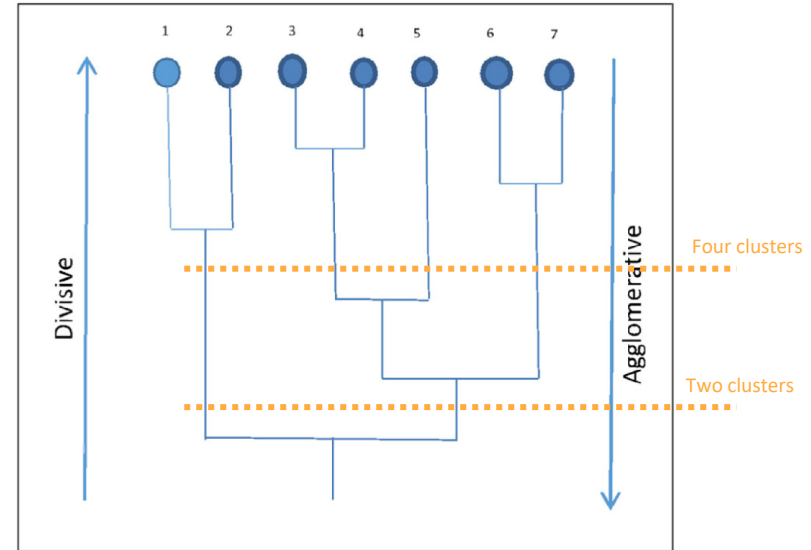


Fig. 2. A dendrogram representation for hierarchical clustering of data objects 1, 2, 3, 4, 5, 6, 7..

3. Clustering Method based on Partitioning – K-Means

- Goal = regard the center of data points as the center of the corresponding cluster, by iterative computation until some criteria for convergence is met
- Advantages: relatively low time complexity and high computing efficiency
- Disadvantages: not suitable for non-convex data, relatively sensitive to the outliers, easily drawn into local optimal, the number of clusters needed to be preset, and the clustering result sensitive to the number of clusters

One of the most popular and simple clustering algorithms, K-means, was first published in 1955.

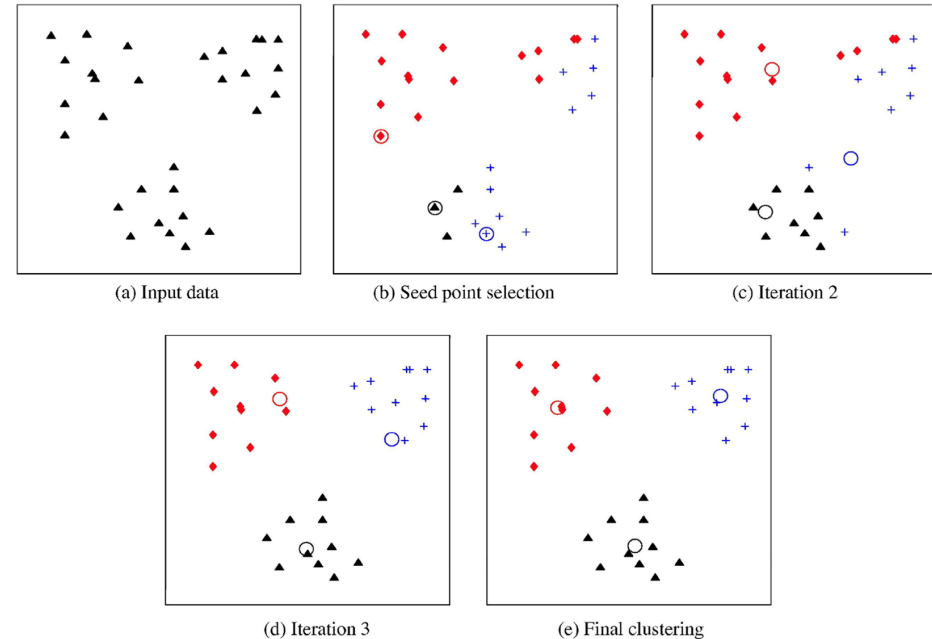
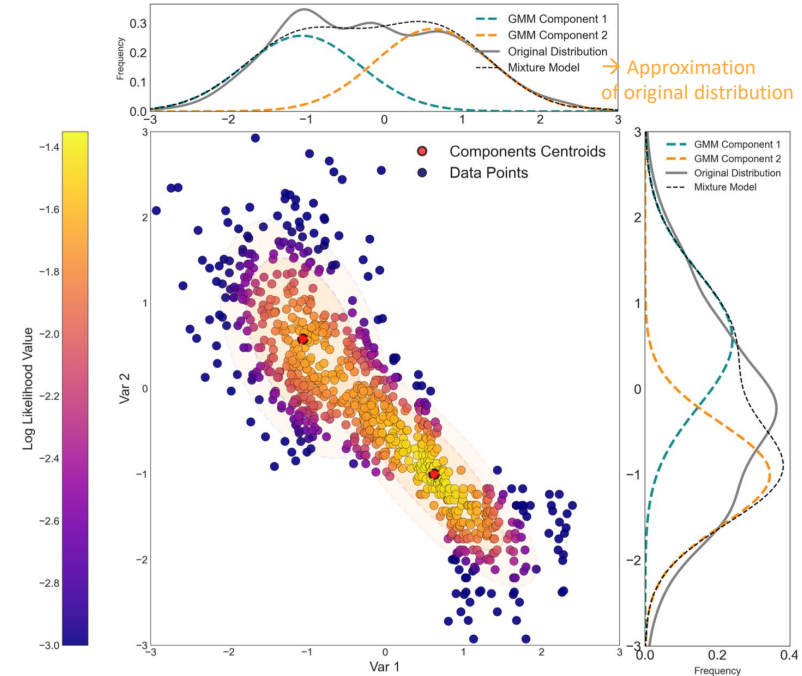


Illustration of K-means algorithm. (a) Two-dimensional input data with three clusters; (b) three seed points selected as cluster centers and initial assignment of the data points to clusters; (c) and (d) intermediate iterations updating cluster labels and their centers; (e) final clustering obtained by K-means algorithm at convergence.

3. Clustering Algorithms based on Distribution – GMM

- basic idea: data, generated from the same distribution, belongs to the same cluster if there exist several distributions in the original data.
- The core idea of GMM (Gaussian Mixture Model): There exist several Gaussian distributions from which the original data is generated and the data, obeying the same independent Gaussian distribution, is considered to belong to the same cluster.
- Advantages: more realistic to give the probability of belonging, high scalability by changing the distribution, number of clusters and so on, and supported by the well developed statistical science
- Disadvantages: the premise not completely correct, involved in many parameters which have a strong influence on the clustering result and relatively high time complexity



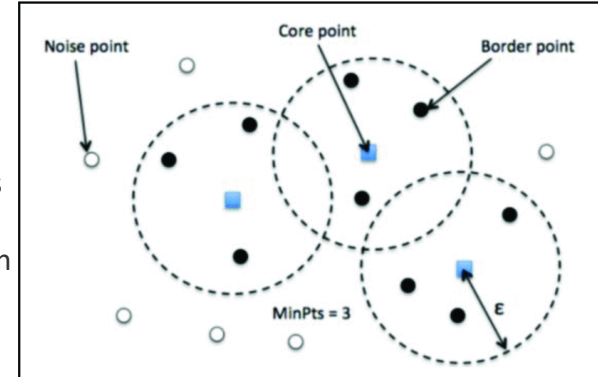
3. Clustering Algorithms based on Density – DBSCAN

- basic idea: data which is in the region with high density of the data space is considered to belong to the same cluster.
- DBSCAN (density-based spatial clustering of applications with noise) is the most well known density-based clustering algorithm
- Advantages: high efficiency and suitable for data with arbitrary shape
- Disadvantages: resulting in a clustering result with low quality when the density of data space isn't even, a memory with big size needed when the data volume is big, and the clustering result highly sensitive to the radius of the neighborhood and the minimum number of points in a neighborhood

Hyper Parameters:

- $\text{eps}(\epsilon)$: - This is the radius around the point which is checked for other points in its proximity. It is used to determine the density of region

- minPts : - This is the number of points present in the surrounding (eps distance from the point) to form a cluster.



Point Classification:

1. Core — If different clusters are made this is the point from which there are at least k number of points in r distance of radius.

2. Border — This can be any point that has one or more core points within r distance of radius.

3. Noise — Any point which is neither a core and a border and has at least k number of points in r distance of radius.

3. Model Selection - Clustering Algorithms

„Each clustering algorithm imposes a structure on the data either explicitly or implicitly. When there is a good match between the model and the data, good partitions are obtained. Since the structure of the data is not known a priori, one needs to try competing and diverse approaches to determine an appropriate algorithm for the clustering task at hand. “

→ impossibility theorem ([Kleinberg, 2002](#)): no single clustering algorithm simultaneously satisfies a set of basic axioms of data clustering

4. Evaluation Indicators for Clustering

- Purpose: test the validity of the clustering algorithm
- two categories:
 - a) **internal evaluation indicators** (measure performance upon data set) → Table 3

An ideal cluster :

„ A cluster is a set of points that is compact and isolated.“

Properties that we want to evaluate for measuring clustering performance:

- (1) *Minimal distances within one cluster*
= compactness
- (2) *Maximal distance across clusters*
= separation

Distance metric matters!

Table 3 Evaluation indicators

Name	Formula or measure method	Explanation
Davies–Bouldin indicator	$DB = \frac{1}{k} \sum_{i=1}^k \max_{i \neq j} \left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j)} \right)$	K stands for the number of clusters, C_x is the center of cluster x , σ_x is the average distance between any data in cluster x and C_x , $d(c_i, c_j)$ is the distance between c_i and c_j
Dunn indicator	$D = \min_{1 \leq i \leq n} \left\{ \min_{1 \leq j \leq n, i \neq j} \left\{ \frac{d(i, j)}{\max_{1 \leq k \leq n} d'(k)} \right\} \right\}$	1. Mainly for the data that has even density and distribution 2. $d(c_i, c_j)$ is the distance between c_i and c_j , $d'(k)$ stands for the distance in cluster k
Silhouette coefficient	Evaluate the clustering result based on the average distance between a data point and other data points in the same cluster and average distance among different clusters	

4. Evaluation Indicators for Clustering

- Purpose: test the validity of the clustering algorithm
- two categories:
 - a) **internal evaluation indicators** (measure performance upon data set) → Table 3
 - b) **external evaluation indicators** (compare against externally given clustering; labels) → Table 4
= measuring accordance to other clustering result

Table 4 Evaluation indicators

Name	Formula or measure method	Explanation
Rand indicator	$RI = \frac{TP+TN}{TP+FP+FN+TN}$	1. TP is the number of true positives 2. TN is the number of true negatives 3. FP is the number of false positives 4. FN is the number of false negatives
F indicator	$F_{\beta} = \frac{(\beta^2+1) \cdot P \cdot R}{\beta^2 \cdot P + R}$	1. $P = \frac{TP}{TP+FP}$ stands for the accuracy, $R = \frac{TP}{TP+FN}$ stands for the recall rate 2. TP, TN, FP, and FN are defined as before
Jaccard indicator	$J(A, B) = \frac{ A \cap B }{ A \cup B } = \frac{TP}{TP+FP+FN}$	1. Measure the similarity of two sets 2. X Stands for the number of elements of set X 3. TP, TN, FP, and FN are defined as before
Fowlkes–Mallows indicator	$FM = \sqrt{\frac{TP}{TP+FP} \cdot \frac{TP}{TP+FN}}$	TP, TN, FP, and FN are defined as before
Mutual information	To measure, based on information theory, how much information is shared by two clusters, between which the nonlinear correlation can be detected	
Confusion matrix	To figure out the difference between a cluster and a gold-standard cluster	

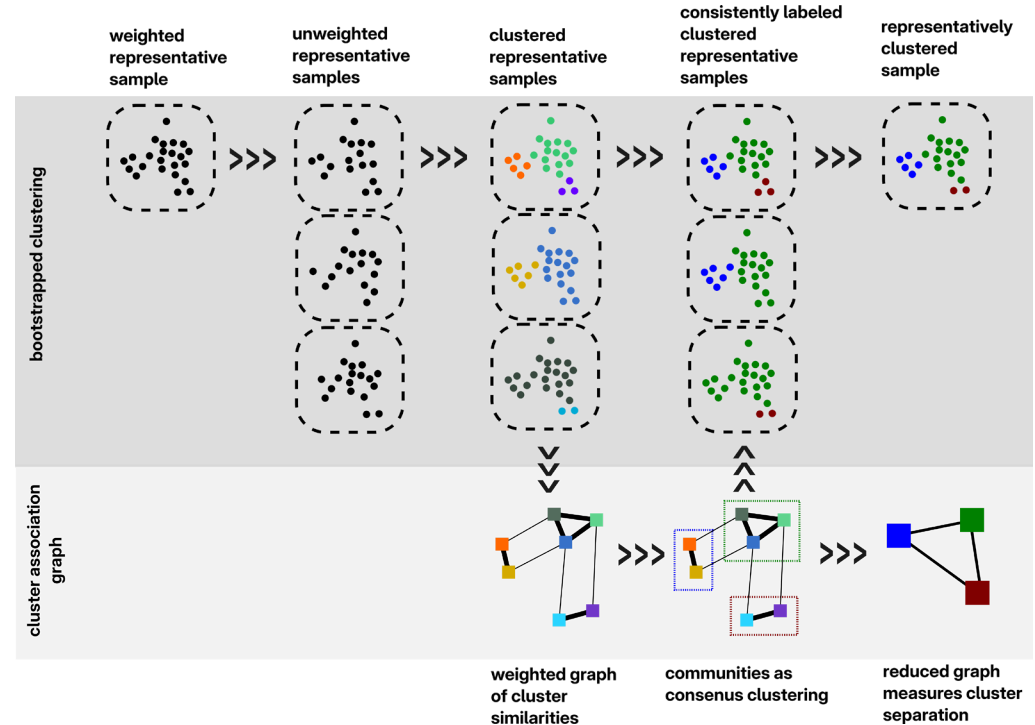
National Representative & Stable Population Segmentation

Aim: health-care utilization-based or needs-based population segmentation

Benefits: Healthcare needs evaluation, outcome tracking, care integration, and service interventions designed in relation to need

Development of new methodology for clustering enables:

- Nationally representativeness
- Robust findings
- Identification of relevant number of cluster



Clustering Multimorbidity - What is meant?

Multimorbidity = the simultaneous occurrence of two or more long-term conditions in an individual.

There is not one right choice for clustering multimorbidity: The various approaches available have different rationales and implications

Different intended purposes of the multimorbidity clusters might lead to different choices of clustering.

1. Understanding mechanisms requires the conditions in the cluster to be associated through an underlying common cause
→ intervention design
2. Predicting adverse outcomes requires the conditions in the cluster to influence each other and the outcome
→ targeting interventions
3. Planning health and social care services requires the conditions to be highly prevalent or highly impactful on individuals or services.
→ planning service utilization and resource allocation

Clustering Multimorbidity - What is meant?

Mechanisms behind co-occurrence of long-term conditions:

- by chance (which could be more important in people older than 65 years, in whom individual conditions usually have the highest prevalence)
- because conditions share a common aetiology (such as smoking or obesity or genetic underpinnings)
- because one condition causes another (such as diabetes and microvascular or macrovascular disease)
- because the treatment for one condition causes another (such as aspirin prescribed for coronary heart disease causing stroke due to cerebral haemorrhage)
- or conditions can also be said to cluster when they each lead to a common adverse outcome or care needs, for example, stroke and Parkinson's disease might be aetiologically unrelated but could still both contribute multiplicatively to functional disability and reduced quality of life

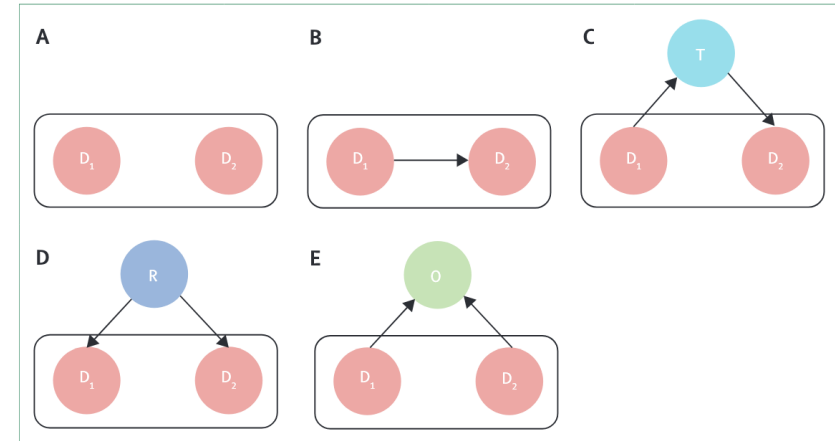


Figure 1: Different situations in which finding clusters of multimorbidity can be of interest

Disease 1 (D₁) and disease 2 (D₂) are a cluster of interest because: they appear frequently by chance (A), D₁ leads to D₂ (B), D₂ is an effect of treating D₁ by treatment (T; C), D₁ and D₂ have a common risk (R; D), or D₁ and D₂ together cause outcome (O; E).

Clustering Multimorbidity - What is meant?

Clustering Multimorbidity choices:

- clustering is applied over conditions (such as asthma, eczema, hay fever) or over individuals (such as people with atopy)
- clustering is done on a cross-section of data (presence or absence of conditions at a moment in time) or in longitudinal data (pattern of condition accrual over time)
- clustering focuses on joint prevalence of conditions (how often they co-occur) or their strength of association (co-occurrences beyond chance)
- clustering focuses on number of conditions, presence of specific conditions, body system relation or effect on quality of life
- clusters are made exclusive (type 2 diabetes is only allowed to be in one condition cluster; or an individual is uniquely allocated to a cluster) or objects can be assigned simultaneously to multiple clusters (type 2 diabetes is allowed to be in clusters of conditions caused by obesity, but also in clusters of conditions that type 2 diabetes causes; an individual can simultaneously be allocated to an atopy cluster and a cardiometabolic cluster).
- Selection of distance metric
- What conditions to include, granularity of condition



Figure 3: A) Exclusive clustering of individuals, while the same condition can be assigned to multiple clusters. B) Exclusive clustering of conditions, while the same individual can be assigned to multiple clusters. C) Non-exclusive assignment of both conditions and individuals.

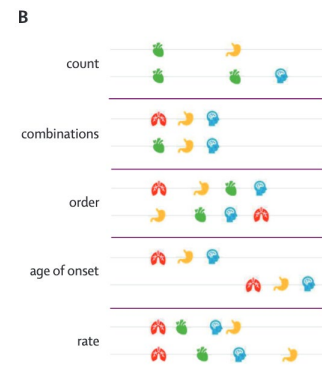


Figure 2 B) Different characteristics of multimorbidity trajectories: the figure illustrates the various ways in which multimorbidity trajectories can differ. In this situation, the trajectories in the later rows differ from the first few rows in terms of the count, combinations, order, onset, and rate of conditions being accrued, and different similarity measures can account for these different facets of the trajectories.

Clustering Multimorbidity - What is meant?

Desired Characteristics of analyses clustering multiple long-term conditions:

1. Stability to perturbation
2. Stability across methods
3. Stability across datasets (reproducibility)
4. Explainability (How clusters are formed and how individuals can be assigned)
5. Fairness
6. Clinical plausibility or utility and mechanistic underpinnings (operationalizability, relevance for downstream analysis e.g. through association with relevant clinical outcomes)

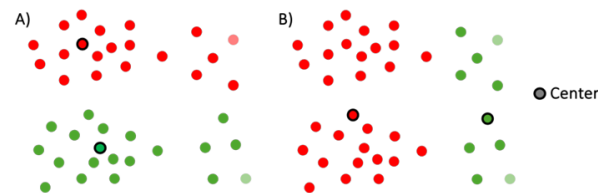


Figure 4: A) Unfair clustering of individuals since an individual (shown as light color) can be far away from the respective cluster centre, B) Fair clustering since every individual is only a certain distance away from the respective cluster centre.

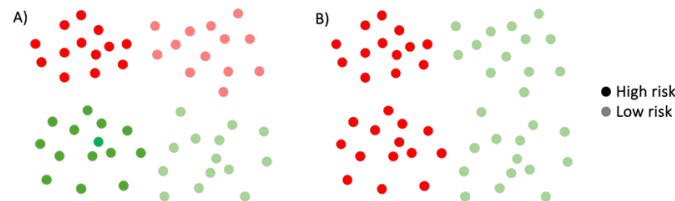
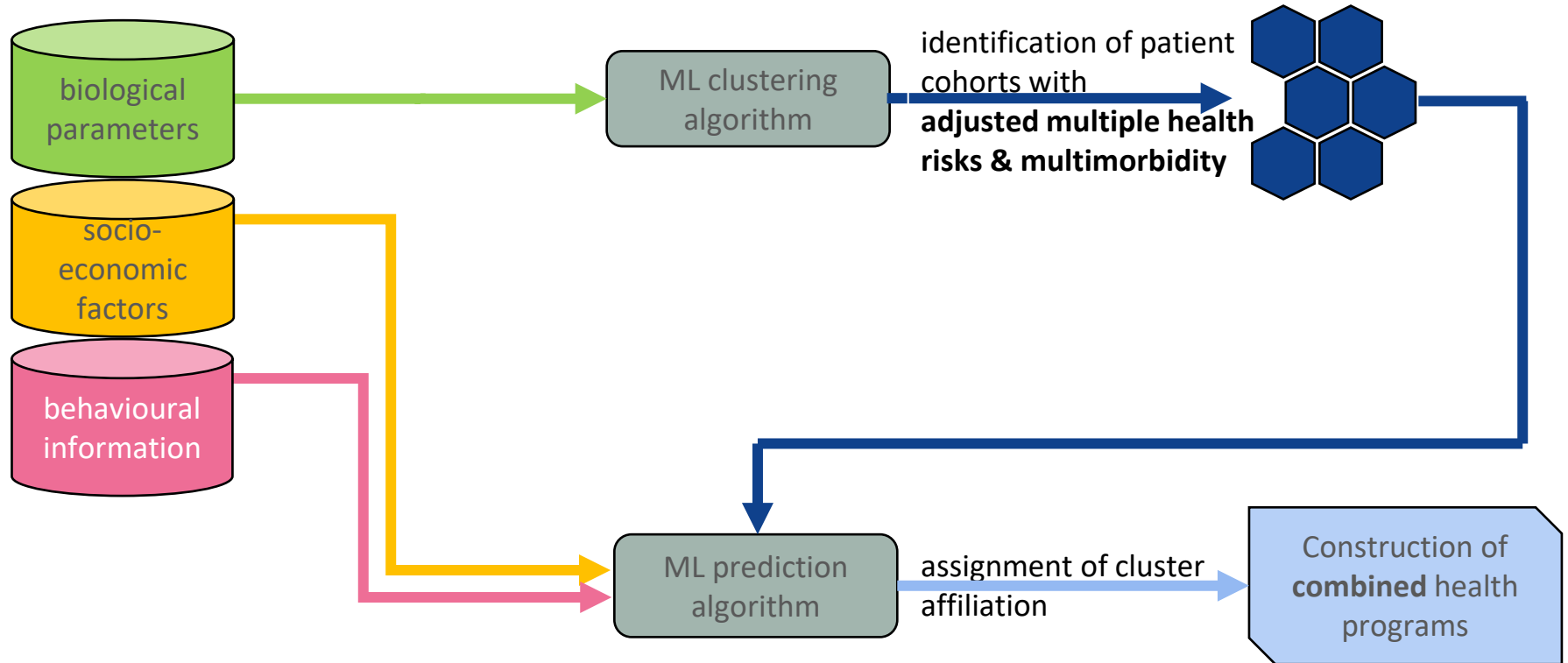


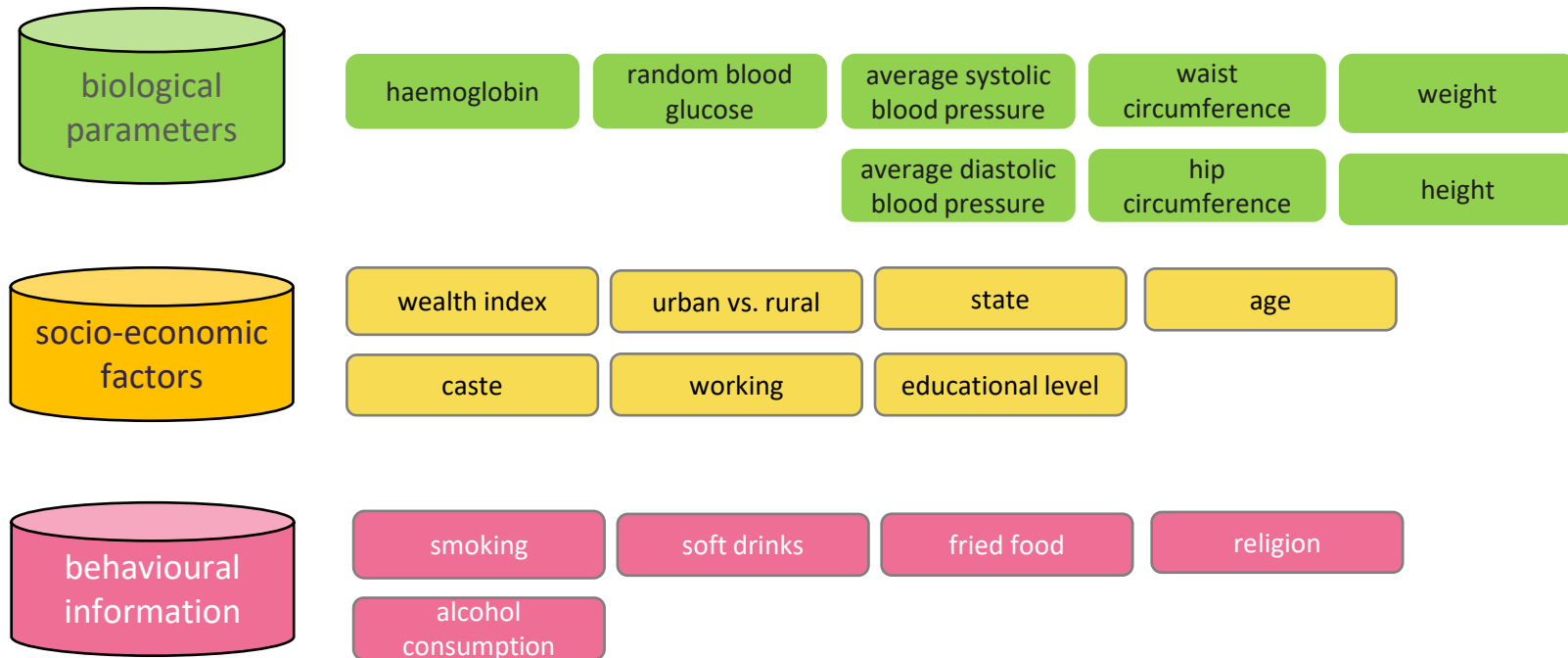
Figure 5: A) Less operationalizable clustering since both high and low risk individuals can belong to the same cluster, B) more operationalizable clustering since it separates the high and low risk individuals.

„Therefore, we encourage moving from a top-down approach of unsupervised clustering of long-term conditions and exploring generic attributes of the clusters to a bottom-up approach of considering the purpose of clustering in a particular context (and using this to guide the choices made during clustering analysis) and reporting and explicitly justifying these choices to improve comparability and reproducibility (and facilitate evaluation of utility for different purposes).“ (Seth et al. 2026)

Clustering Analysis



Variable Selection



Method

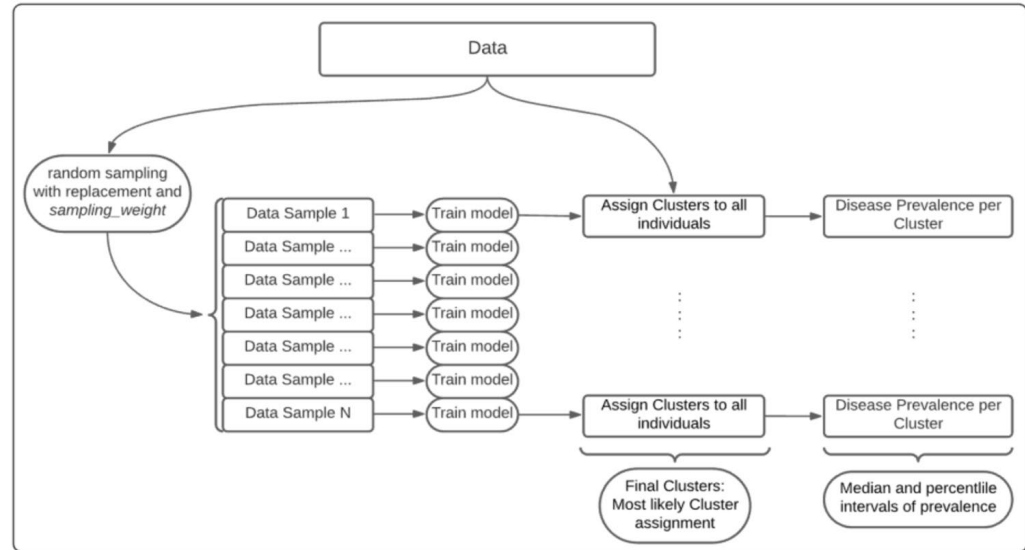
Data Preprocessing:

- defined biologically realistic limits and implausible value combinations
- of random blood glucose and blood pressure required further processing to account for altering factors, such as medication
- Scaling to a mean of 0 and a standard deviation of 1

Clustering Analysis:

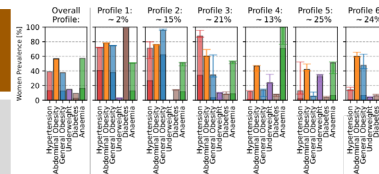
- Method = k-means
- 1000-times bootstrapping procedure by sampling with replacement individuals out of the dataset proportional to the weights, accounting for the survey design
- Cluster labels were aligned across the runs by calculating the largest share of clusters' individuals between the runs
- number of clusters k was determined through a best balance between statistical stability, interpretability, and policy relevance
- Description of cluster characteristics, including the number of cooccurring diseases, disease prevalences, wealth index factor score, age, and geography

Clustering with repeated sampling

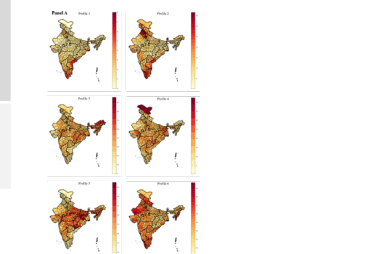
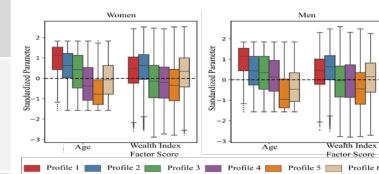


Results

	Suggested profile label	Weighted share among women, % (95% CI)	Weighted share among men, % (95% CI)	Dominant conditions, diagnostic threshold	Median age, women/men	Median wealth score, women/men	Main geographic concentration	Programme implication
Profile 1	High multimorbidity, diabetes profile	2.2 (2.2 – 2.2)	2.6 (2.3 – 3.0)	Obesity, hypertension, diabetes	42/42	6.22/6.01	Southern and north-eastern India; no district >50%	Integrated diabetes-hypertension-obesity care
Profile 2	High multimorbidity, non-diabetes profile	14.7 (11.7 – 15.0)	17.5 (12.58 – 19.2)	Obesity, hypertension, diabetes lower than Profile 1	37/35	7.68/7.77	Northern states and southern coastline	Obesity and hypertension control; diabetes prevention
Profile 3	Moderate multimorbidity, hypertension profile	21.2 (15.6 – 22.0)	18.4 (17.4 – 22.4)	Hypertension dominant	35/34	0.20/1.40	North-east; west/east among men	Community hypertension detection and affordable treatment
Profile 4	Moderate multimorbidity, non-hypertension, anaemia profile	13.5 (13.1 – 21.6)	18.8 (15.7 – 26.5)	Anaemia dominant	27/32	-0.31/1.80	Ladakh, around 40%; also east/west/south/north-east/central India	Integrate anaemia, nutrition, and NCD care
Profile 5	Low multimorbidity, underweight profile	24.7 (20.7 – 25.0)	20.8 (17.6 – 22.1)	Underweight; low cardiometabolic burden	23/21	-1.83/-2.38	North-east/east/central India; broader among men	Nutrition-sensitive prevention and life-course health
Profile 6	Low multimorbidity, non-underweight profile	23.7 (22.4 – 26.1)	22.1 (20.4 – 25.8)	Low multimorbidity; no underweight	30/26	4.97/3.19	Southern, northern, north-eastern India; broader in men	Maintain low-risk status; prevent future cardiometabolic risk



		Women					
		Profile 1	Profile 2	Profile 3	Profile 4	Profile 5	Profile 6
Number of Districts	5	2.1 (2.0 - 2.3)	4.1 (3.3 - 4.6)	1.4 (1.2 - 3.0)	0.2 (0.0 - 0.3)	0.0 (0.0 - 0.3)	0.1 (0.1 - 0.4)
	4	4.1 (4.0 - 4.3)	2.1 (2.2 - 2.9)	1.4 (1.2 - 2.3)	1.4 (0.8 - 4.2)	1.4 (0.6 - 5.9)	1.2 (0.6 - 6.8)
	3	2.6 (2.5 - 2.8)	4.1 (4.0 - 4.3)	3.9 (3.3 - 4.1)	2.9 (1.2 - 2.3)	1.4 (0.6 - 5.9)	1.2 (0.6 - 6.8)
	2	7.2 (7.6 - 7.7)	2.1 (1.9 - 2.5)	4.1 (2.6 - 3.5)	6.9 (4.8 - 46.9)	2.7 (2.7 - 3.9)	3.0 (3.9 - 40.2)
1	5	1.5 (1.5 - 1.5)	2.4 (2.9 - 5.3)	1.2 (6.0 - 14.5)	7.8 (26.1 - 16.3)	37.3 (26.7 - 43.2)	2.1 (20.7 - 29.7)
	4	0.0 (0.0 - 0.0)	0.1 (0.0 - 0.1)	1.1 (0.2 - 1.5)	0.0 (0.2 - 8.0)	13.6 (7.9 - 20.1)	9.0 (3.0 - 9.3)
	3						
	2						



Unsupervised Machine Learning

Impossibility theorem (Kleinberg, 2002): No single clustering algorithm simultaneously satisfies a set of basic axioms of data clustering. Competing and diverse approaches to determine an appropriate algorithm for the clustering task at hand need to be performed.

Population Segmentation

Derivation of new clustering workflow: including the statistical representation of survey designs to achieve national representativeness; enabling the assessment of the clustering confidence; as well as allowing for negative clustering results (no hyperparameter k)

Multimorbidity-based Population Segmentation

There is not one right choice for clustering multimorbidity: The various approaches available have different rationales and implications. The purpose of clustering in a particular context should lead the choices of clustering. Reporting and explicitly justifying these choices should improve comparability and reproducibility.

Use Case: Clustering Multimorbidity on NFHS-5 India

On biological biomarkers six representative population segmentation could be derived for women and men in India based on NFHS-5.

Finding groups and patterns in large datasets

Clustering intuition and practical examples.

Suitable questions, preprocessing, cluster stability, and interpretation.



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Variable Selection

